

CS 544 (Spring 2026): Project 3 Report

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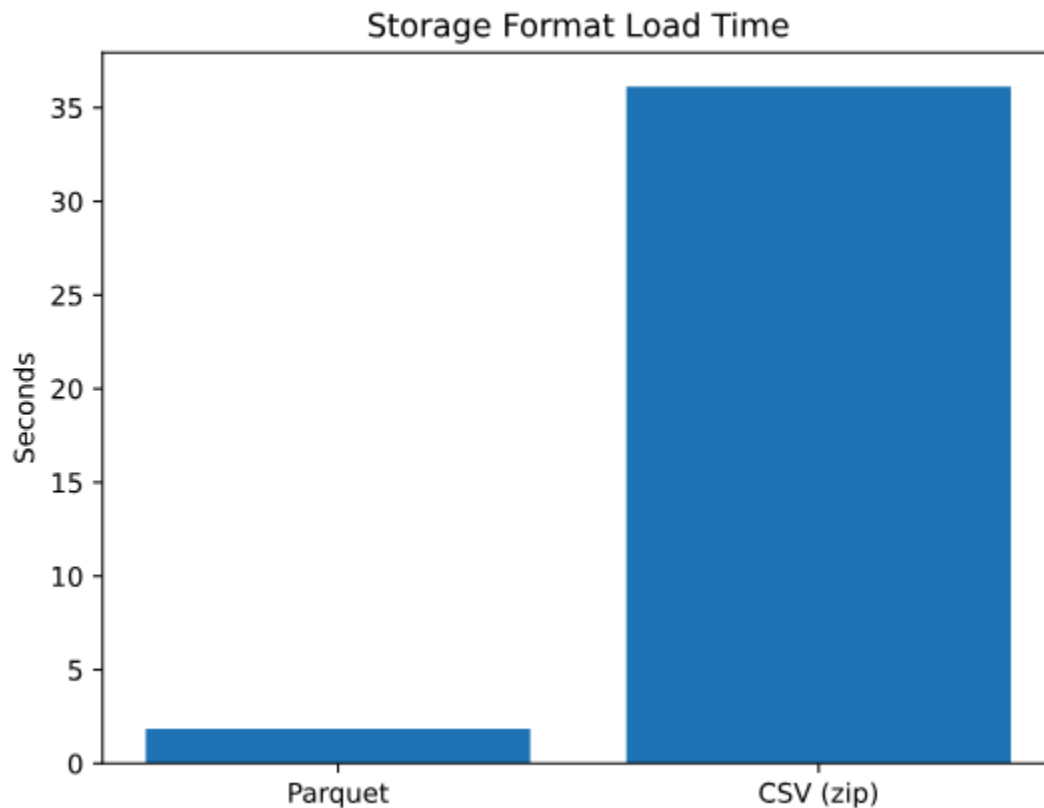
Don't change the number of pages, and don't let content for a specific part exceed the assigned page.

Part 2: Storage Performance

In your own words, what question are we asking with this experiment?

Is column-oriented (parquet) or row-oriented (zipped CSV) parsing faster?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

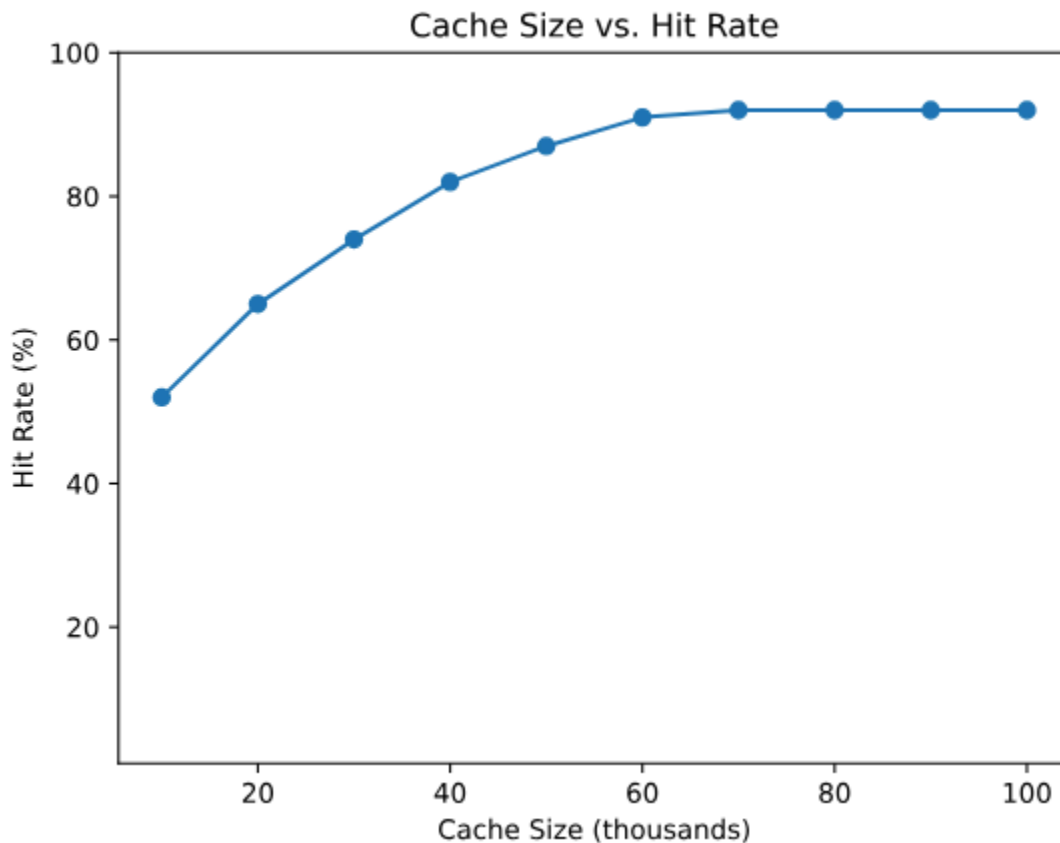
Parquet with ~2 seconds is faster than zipped CSV with ~35 seconds by 17.5 times

Part 3: Memory Performance

In your own words, what question are we asking with this experiment?

What is the relationship between cache size and hit rate?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

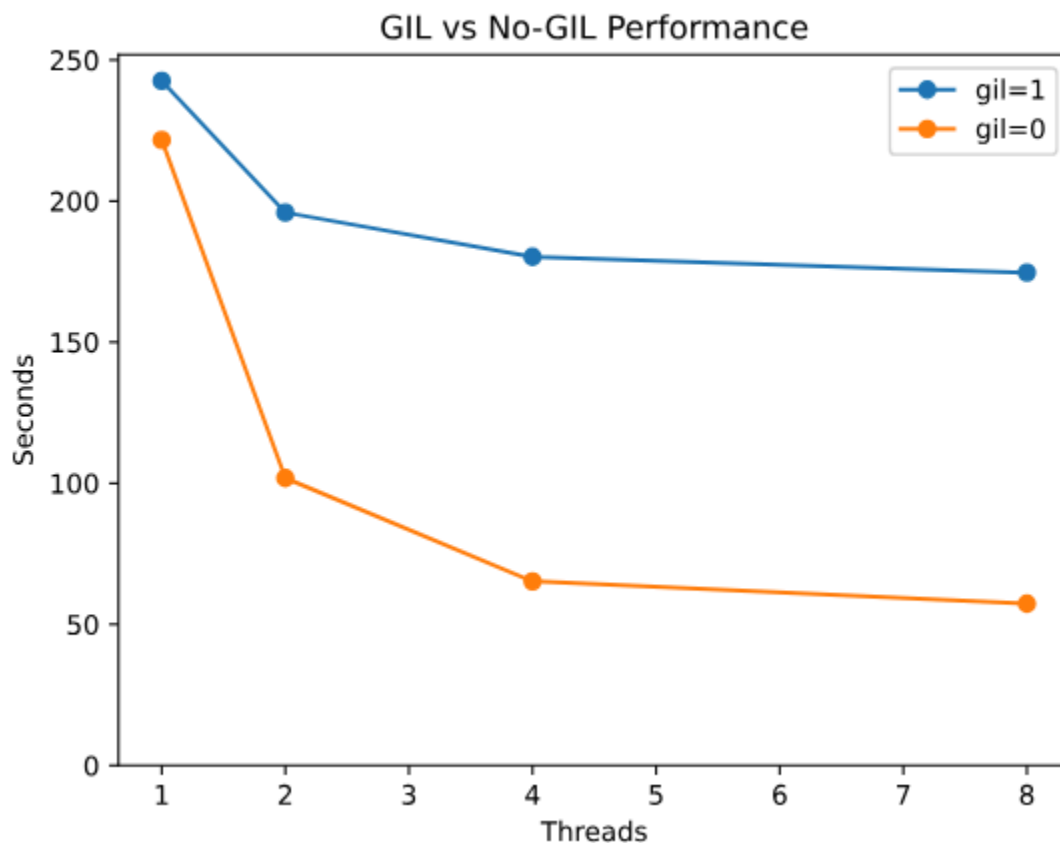
As cache size increases, hit rate also increases. However at a certain point of cache size (~50k), the hit rate begins to plateau at ~92%.

Part 4: Compute Performance

In your own words, what question are we asking with this experiment?

How does the computation time of GIL and No-GIL change with varying numbers of threads?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

When the number of threads are fewer (1), GIL and No-GIL perform about the same. However, as threads increase (8), GIL shows slower computation time (~175 seconds) compared to No-GIL (~75 seconds).